

Republic of the Philippines
CATANDUANES STATE UNIVERSITY
Virac, Catanduanes

Activity 2
Math 408 — Advanced Calculus II

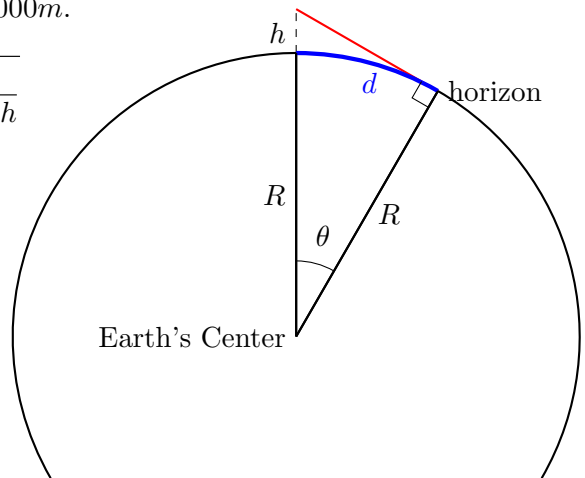
Name: _____
Student Number: _____
Program and Block: _____

Total Score: _____
Date: _____
Instructor: William Angelo V. Gonzales

This activity contains 3 sections. Total of points is 25.
Answer the questions independently.

1. (10 points) **Distance to the Horizon.** The formula below approximates the curved ground distance of the earth d to the horizon where h is the elevation from the sea level and R is the radius of the earth. For the purpose of this activity, we will use $R = 6,371km$ or $6,371,000m$.

$$d \approx R\sqrt{\frac{2h}{R+h}}$$



- (a) (1 point) Compute the curved ground distance to the horizon when $h = 5m$. _____
- (b) (1 point) Compute the curved ground distance to the horizon when $h = 10m$. _____
- (c) (1 point) Compute the curved ground distance to the horizon when $h = 20m$. _____
- (d) (1 point) Compute the curved ground distance to the horizon when $h = 30m$. _____
- (e) (2 points) The angle θ in the figure is in radians. Write a formula for d in terms of the radius R and the angle θ . _____
- (f) (2 points) Notice that the triangle formed is a right triangle, use trigonometry to find a formula for d in terms of R and h . _____
- (g) (2 points) Using your formula in the previous item, compute the curved ground distance when $h = 5m$, $h = 10m$, $h = 20m$ and $h = 30m$. Compare this with your answers in 1(a) through 1(d). What do you observe?

2. (3 points) **Geometric Series.** Consider this function which can be expressed as an infinite series:

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + x^5 + \dots$$

- (a) (1 point) For what values of x does this infinite series converge?
- (b) (1 point) Will the series converge when $x = 1$? Why?
- (c) (1 point) Will the series converge when $x = -1$? Why?

3. (12 points) **Integration of an Infinite Series.**

- (a) (1 point) Find an infinite series for
- $\frac{1}{1+x}$
- .

$$\frac{1}{1+x} =$$

- (b) (2 points) For what values of
- x
- does this infinite series converge? Explain.

- (c) (2 points) Integrate both sides of the equation in 3(a) with respect to
- x
- . Don't forget to put "+
- C
- " at the end of the infinite series after integration.

$$\int \frac{1}{1+x} dx = \int \left(\right) dx$$

$$=$$

- (d) (1 point) Set
- $x = 0$
- to the resulting equation after integration and solve for
- C
- .

- (e) (1 point) Write the new infinite series below, substituting the computed value for
- C
- .

- (f) (2 points) Will this new infinite series converge for all real values for
- x
- ? To answer this, use the
- partially completed*
- python code below. Line 2 is where you put a test value for
- x
- , and line 5 is the
- n
- th term.

Python 3 Console	
1	<code>import math</code>
2	<code>x =</code>
3	<code>sum = 0</code>
4	<code>for n in range(1000):</code>
5	<code> a = (((-1)**(n))*(x**(n+1)))/(n+1)</code>
6	<code> sum += a</code>
7	<code>print(sum)</code>

Give the interval for x that causes the infinite series to converge: $\{x \in \mathbb{R} \mid \quad \quad \quad \}$

- (g) (2 points) In this activity, we were able to come up with an entirely different infinite series by integrating another infinite series. Do these series have the same interval of convergence? If not, what is the difference in their intervals of convergence?

- (h) (1 point) Give the value of the infinite series below:

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \frac{1}{7} - \frac{1}{8} + \frac{1}{9} - \cdots =$$